

Learning Intention: I can read and create picture graphs and column graphs to display data.

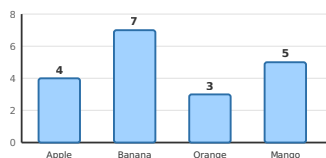
Success Criteria — I can:

- Read information from a picture graph and a column graph.
- Create a graph with a title, labels and a key or scale.
- Compare two graphs and describe what I notice.

Strategy Box

Before reading any graph, check three things: **1) the title** — what is the graph about? **2) the labels** — what do the axes show? **3) the key or scale** — what does each picture or grid line represent? Then read the values carefully.

Worked Example — Favourite Fruit Survey



1. **Title:** "Favourite Fruit Survey" tells us this graph is about fruit choices.
2. **Scale:** the side numbers count up in 2s (0, 2, 4, 6, 8).
3. The tallest bar is **Banana** with 7 votes — that was the most popular.
4. The shortest bar is **Orange** with 3 votes — that was the least popular.

Review: Parts of a Graph

Part	What it does
Title	Says what the graph shows
Labels	Name what each axis measures
Key / Scale	Shows what one picture or grid line equals

Picture Graph vs Column Graph

A **picture graph** uses repeated pictures or symbols, each worth a set amount (shown in a key). A **column graph** uses bars measured against a number scale. Both can show the same data in different ways.

Try It — Read the Worked Graph

- a) How many people chose Apple? _____ b) How many more people chose Banana than Orange? _____
 c) What is the total number of people surveyed? _____

Parent Tip

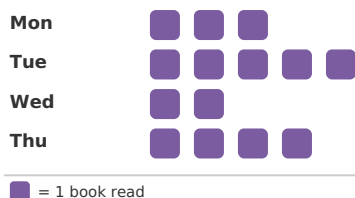
Look at a graph together next time you see one — in the news, a recipe book, or a sports website. Ask your child: "What is this graph about? What is the biggest number shown?"

Key Vocabulary

Word	Meaning
Data	Information collected, often as numbers or counts
Survey	Asking a group of people the same question to collect data
Scale	The numbers along the side of a column graph
Key	Explains what one picture is worth on a picture graph

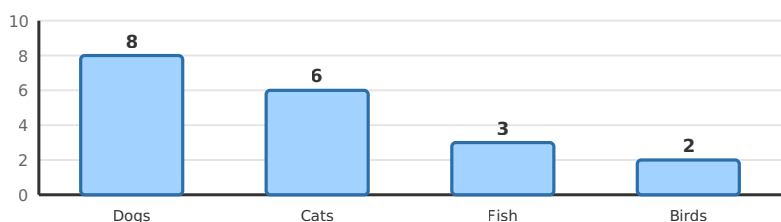
Build Skills — Read, create and compare graphs.

Q1 — Read the Picture Graph: Books Read This Week



- a) How many books were read on Tuesday? _____ b) Which day had the fewest books read? _____
 c) How many more books on Tuesday than Wednesday? _____ d) Total books read across all 4 days? _____

Q2 — Read the Column Graph: Pets Owned by Our Class

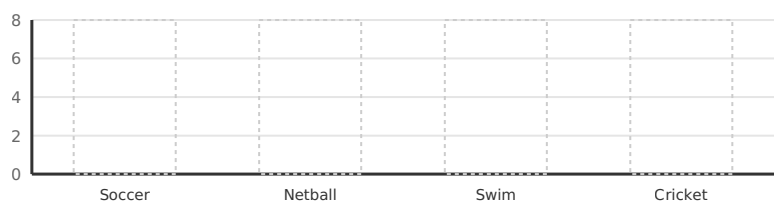


- a) Which pet is owned the most? _____ b) How many more dogs than birds? _____ c) Total pets shown? _____

Q3 — Create Your Own Column Graph

Use the data table to draw bars on the blank graph. Colour each bar up to the correct number.

Sport	Soccer	Netball	Swimming	Cricket
Votes	6	4	7	3



Watch Out!

Always check the **scale** before reading a value off a graph — if the grid lines count up in 2s, a bar sitting one line above 4 means **6**, not 5. Never guess a value just by looking at bar height alone.

Apply & Reasoning — Use what you know to explain your thinking.

- 1 Reasoning:** Look at the Pets Owned column graph. Would a picture graph or a column graph be better for showing 100 pets? Explain your thinking.

- 2 Reasoning:** Why does a graph need a title and labels? What could go wrong if they were missing?

- 3 Open-ended:** Write a survey question you could ask your class (e.g. favourite colour, favourite season). Predict what the data might show and explain why.

Quick Check — Exit Ticket

Look back at the Books Read picture graph on Page 2. Which day had the most books read, and how do you know?

Answer & Teacher Support

Answer Key — Pages 1 & 2

Try It (p1): a) 4 people chose Apple. b) 4 more chose Banana than Orange (7–3). c) Total = $4+7+3+5 = 19$ people.

Q1 (p2): a) 5 books. b) Wednesday (2 books). c) 3 more (5–2). d) Total = $3+5+2+4 = 14$ books.

Q2 (p2): a) Dogs (8). b) 6 more (8–2). c) Total = $8+6+3+2 = 19$ pets.

Q3 (p2): Bars should reach: Soccer 6, Netball 4, Swimming 7, Cricket 3, matching the data table exactly.

Answer Key — Page 3

Q1: A column graph (with a larger scale, e.g. counting in 10s) suits big numbers better — a picture graph with 100 pets would need too many symbols.

Q2: Without a title/labels, the reader can't tell what the data is about or what is being measured — the graph becomes meaningless.

Q3 (acceptable-answer guide): Accept any sensible survey question with a prediction that includes a reason (e.g. "I think blue will win because it's a popular colour at our school").

Exit ticket: Tuesday had the most books read (5) — it has the tallest bar / most icons.

Teacher Checklist

- ☐ Reads values from a picture graph using the key.
- ☐ Reads values from a column graph using the scale.
- ☐ Draws bars accurately from a data table.
- ☐ Compares two graph types and explains a preference.
- ☐ Explains the purpose of a title, labels and key/scale.

Teacher Tip

Watch for students reading bar height without checking the scale — a common error when the scale counts in 2s or 5s rather than 1s. Model "reading across" from the bar top to the y-axis explicitly.